Mississippi Valley State University

CS 341 - Discrete Structures

Practice Set 2

Negation, Conjunction, Disjunction, and Truth Tables

Week 1 - Lecture 2 - Estimated time: 45-60 minutes

Name: _____ **Date:** _____

How to use this packet

1. Complete the first attempt without the completed notes or approved solutions.
2. Show intermediate columns in every truth table; do not jump directly to the final column.
3. Mark a confidence score after each section.
4. After checking an approved solution, correct each error in a different color and write what caused it.

This is ungraded practice. The goal is to make your reasoning visible enough that you can find and repair a mistake.

A focused study routine

Time	Step	What to do
5 minutes	Recall	From memory, write the symbols for not, and, inclusive or, and exclusive or.
10 minutes	Structure	Complete Problems 1-5. Check row counts and standard row order.
10 minutes	Connectives	Complete Problems 6-18. Say the truth condition aloud before filling an entry.
10 minutes	Translate	Complete Problems 19-24. Preserve parentheses and the scope of negation.
15 minutes	Tables and code	Complete Problems 25-30. Use intermediate columns and test every required case.
5 minutes	Repair	Check work, correct errors, and record one question.

Mastery checklist

Place a check only when you can do the task and explain why.

- ☐ I can determine how many rows a complete truth table needs.
- ☐ I can negate a statement without dropping a boundary case.
- ☐ I know exactly when $\wedge$, $\vee$, and $\oplus$ are true.
- ☐ I can translate between English, symbols, and code.
- ☐ I can evaluate a compound statement one operation at a time.
- ☐ I can use a truth table to expose a faulty program condition.

1 Truth-Table Foundations

1. Row counts. Complete the table. Then state the general rule for n component statement variables.

Variables	1	2	3	5
Rows required				

General rule:

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2. Standard order for two variables. Fill in the missing entries.

P	Q
T	T
T	___
___	T
F	___

3. Standard order for three variables. Complete the eight input rows. Change R every row, Q every two rows, and P every four rows.

P	Q	R	P	Q	R
T	T	T	F	T	T
T	T	___	F	T	___
T	___	T	F	___	T
T	___	___	F	___	___

4. Explain the count. Why does a truth table with four variables require 2^4 rows? Connect your explanation to the multiplication principle or to binary strings.

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5. Find the defect. A student proposes the following complete input table for two variables:

P	Q
T	T
T	F
F	F
T	T

Identify the duplicated row and the missing row. Explain why this table cannot verify a compound statement for every case.

Confidence: 1 = unsure 2 = developing 3 = ready to explain

2 Negation

6. Complete the truth table for negation.

P	$\sim P$
T	
F	

For Problems 7-11, write the logical negation in the clearest ordinary language or mathematical notation. Include all boundary cases.

7. P: "The database is available."

8. Q: "The file contains at least 20 records."

9. $R(x)$: $x \leq 7$.

10. $S(n)$: "The integer n is even."

11. T: "Exactly one process is running."

12. Error analysis. A student claims that the negation of $x < 5$ is $x > 5$.

(a) Give a value of x for which both the original statement and the student's proposed negation are false.

(b) State the correct negation.

(c) Explain which boundary case the student lost.

Scope warning

In $\sim P \vee Q$, only P is negated. In $\sim (P \vee Q)$, the entire disjunction is negated. Parentheses are part of the meaning, not decoration.

Quick scope check. Underline the complete expression affected by $\sim$ in each formula:

$$\sim P \wedge Q \qquad \sim (P \wedge Q) \qquad P \vee (\sim Q)$$

Confidence: 1 = unsure 2 = developing 3 = ready to explain

3 Conjunction, Disjunction, and Exclusive Or

13. Complete all three output columns.

P	Q	$P \wedge Q$	$P \vee Q$	$P \oplus Q$
T	T			
T	F			
F	T			
F	F			

14. Evaluate a system state. Let

P : "The server is online." Q : "The backup is current."

Suppose $P = T$ and $Q = F$. Determine the truth value of each statement and explain it in the system context.

Expression	Value	Contextual explanation
$P \wedge Q$		
$P \vee Q$		
$P \oplus Q$		

15. Inclusive or exclusive? Write $\vee$ or $\oplus$ for the intended meaning.

(a) A search result appears if the keyword occurs in the title or body; both are allowed. _____

(b) Select exactly one authentication method for this step: SMS or security key. _____

(c) A student may use a laptop or tablet during the activity, including both. _____

(d) A bit is either 0 or 1, but not both. _____

16. Explain why the first row of the truth table for $P \vee Q$ is T, while the first row for $P \oplus Q$ is F.

17. Complete the sentences.

(a) $P \wedge Q$ is true only when _____.

(b) $P \vee Q$ is false only when _____.

(c) $P \oplus Q$ is true only when _____.

18. Create. Write one realistic computing example of an inclusive-or rule and one example of an exclusive-or rule. Make the intended meaning unambiguous.

Confidence: 1 = unsure 2 = developing 3 = ready to explain

4 Translate Between English and Symbols

For Problems 19-24, let

$$P: \text{ “The program compiles.”} \quad Q: \text{ “The tests pass.”} \quad R: \text{ “The release is deployed.”}$$

Translate each English statement into symbols.

19. The program compiles and the tests pass.

20. The program compiles, but the tests do not pass.

21. At least one is true: the tests pass or the release is deployed.

22. Exactly one is true: the program compiles or the tests pass.

Translate each symbolic statement into clear English. Preserve the parentheses.

23. $(P \wedge Q) \vee R$

24. $P \wedge \sim (Q \vee R)$

Translation check

After translating, point to the English phrase represented by each connective. If you cannot identify the phrase controlled by a symbol, the translation probably needs revision.

Additional translation drill. These are not separately numbered; use them if you need another attempt.

(a) Write $\sim P \vee (Q \wedge R)$ in ordinary language.

(b) Symbolize: “The program does not compile, or both the tests pass and the release is deployed.”

(c) Explain in words how $\sim P \vee Q$ differs from $\sim (P \vee Q)$.

Confidence: 1 = unsure 2 = developing 3 = ready to explain

5 Evaluate and Build Truth Tables

25. Evaluate from assigned truth values. Suppose $P = T$, $Q = F$, and $R = T$. Show the order of evaluation for at least two expressions.

Expression	Value	Expression	Value
$\sim P$		$P \wedge Q$	
$\sim Q$		$P \vee Q$	
$P \wedge (\sim Q)$		$(P \vee Q) \wedge R$	
$\sim (P \vee Q)$		$P \oplus R$	

Show two evaluation chains.

26. Build the truth table for $P \wedge (\sim Q)$. Complete the intermediate column before the final column.

P	Q	$\sim Q$	$P \wedge (\sim Q)$
T	T		
T	F		
F	T		
F	F		

In one sentence, describe exactly when this expression is true.

27. Build the truth table for $(\sim P) \vee Q$.

P	Q	$\sim P$	$(\sim P) \vee Q$
T	T		
T	F		
F	T		
F	F		

Which input row makes the final expression false? _____

Confidence: 1 = unsure 2 = developing 3 = ready to explain

6 Three-Variable Truth Table

28. Build the complete truth table for $(P \vee Q) \wedge (\sim R)$.

1. Fill the eight standard input rows.
2. Compute $P \vee Q$ and $\sim R$ separately.
3. Use those two intermediate columns to compute the final column.

P	Q	R	$P \vee Q$	$\sim R$	$(P \vee Q) \wedge (\sim R)$
T	T	T			
T	T	F			
T	F	T			
T	F	F			
F	T	T			
F	T	F			
F	F	T			
F	F	F			

Interpret the final column.

(a) How many rows make the expression true? _____

(b) What must be true about R whenever the expression is true?

(c) What must be true about P and Q whenever the expression is true?

(d) Write a computing scenario for P, Q, and R in which this expression would be a sensible condition.

Method check

If your final answers are wrong but your intermediate columns are visible, you can locate the exact operation that caused the error. That is why the method matters as much as the final column.

Confidence: 1 = unsure 2 = developing 3 = ready to explain

7 Logic in Code

29. Find and repair a security bug. The intended rule is:

> Grant access only when the password is correct and the account is not locked.

A developer writes:

```
if password_correct or not account_locked:
    grant_access()
```

(a) Let P mean `password_correct` and L mean `account_locked`. Write the faulty condition in mathematical symbols.

(b) Give two different input assignments for which the faulty code grants access but the intended rule denies it.

(c) Rewrite the Python condition correctly.

(d) Explain why testing only the case $P = T$, $L = F$ would fail to reveal the bug.

30. Design and test a condition. Define three Boolean variables for a realistic computing situation. Write a condition that uses negation, conjunction, and disjunction at least once each.

(a) Define the three variables in words.

(b) Write the condition in mathematical notation using $\sim$, $\wedge$, and $\vee$.

(c) Write the same condition in Python or C/C++/Java notation.

(d) Choose two different input assignments and evaluate the condition for each.

Final self-check

1. The connective I understand best is _____.
2. The error I am most likely to make is _____.
3. One question I will bring to class or office hours is:

Reading connection

This practice set covers Chartrand and Zhang, Section 1.2, pp. 13-18, and *Mathematics for Computer Science*, Section 3.1.1, pp. 42-43. Formal logical equivalence and De Morgan's Laws are intentionally reserved for the next lecture.